Cryptography

6COSC019W- Cyber Security

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Overview

1. Fundamentals of Cryptography
2. Cryptographic Systems
3. Messages Digest and Hashing Functions
4. Cryptanalysis

Introduction and Definitions

Cryptography

Cryptography is a method of storing and transmitting data in a form that only those it is intended for can read and process.

Cryptographic systems

Cryptographic systems are a combination of techniques and algorithms that use specific mathematical principles and computations to provide one or more security objectives and goals.

Cryptanalysis

Cryptanalysis are techniques used for deciphering a message without any knowledge of the enciphering details.

Cryptographic process terms

- **Plaintext**—Information in its original and understandable form.
- **Encryption**—The process of scrambling plaintext into ciphertext (or Enciphering).
- **Ciphertext**—Information in scrambled form.
- **Decryption**—The process of unscrambling ciphertext into plaintext (or Deciphering).

Cryptography in Practice

- Cryptography is much more than encryption
  - Hiding data (Steganography)
  - Resisting forgery (Integrity)
  - Preventing unauthorised access (Confidentiality)
  - Resisting repudiation (Non-Repudiation)
  - Enforcing policy (Policies)
  - Exchanging keys (Keys distributions and exchange)

Data states Methods

Data in Transit/Motion

- Protecting data in transit, or when being transported across a network from the risk of interception by encrypting it.

Data at Rest

- Protecting data at rest such as when stored on a hard drive by the use of encryption even when not in use.

Data in Use/Processing

- Data in use, such as when being processed in RAM or CPU registers, is still data that requires protection , and in modern secure systems, this data can be encrypted.

Goals of cryptography

- Security practitioners use cryptographic systems to meet four fundamental goals:
  - Confidentiality
  - Integrity
  - Authentication
  - Non-Repudiation
- Achieving each of these goals requires the satisfaction of a number of design requirements, and not all cryptosystems are intended to achieve all four goals.

Confidentiality

- Confidentiality in cryptosystems is provided by scrambling the data (Encryption) in a way that can only be read by authorised entity, such as a receiver having the key to descramble (decrypt).
- Secrecy of data is ensured by:
  - Strength of the algorithms
  - Length of the keys
- When Confidentiality is needed in a cryptographic system , we must think about two different types of data:
  - **Data at rest** is that which resides in a permanent location awaiting access.
  - **Data in transit,** or data (in motion or on the wire), is data being transmitted across a network between two systems.

Integrity

- When a message is sent, both the sender and recipient need to know that the message was not altered in transmission
- **Integrity** ensures that data is not altered without authorisation.
  - Message integrity is crucial to message security
  - Check if message received is identical to the sent message.
  - Ensure that stored data was not altered between the time it was created and the time it was accessed.
  - Important for legal contracts
- **Message integrity** is enforced through the use of encrypted message digests, known as **digital signatures** created upon transmission of a message

Authentication in cryptosystems

- **Authentication** verifies the claimed identity of system users and is a major function of cryptosystems
- Matching of a user to an account through previously shared credentials
  - A combination of cryptographic methods are commonly employed in authentication

Non-Repudiation

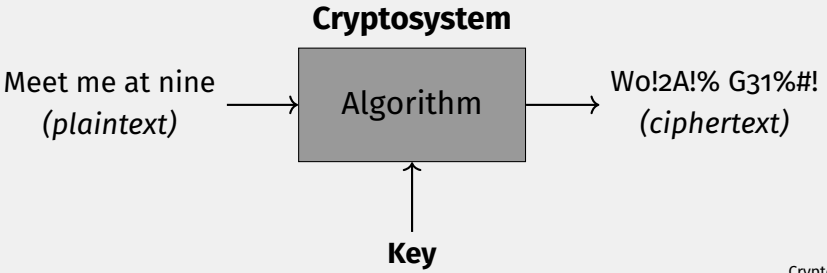
- **Non-Repudiation** provides assurance to the recipient that the message was originated by the sender and not someone masquerading as the sender.
- The message sender cannot later deny that they sent the message
  - Important in electronic exchanges of data because of the lack of face-to-face meetings
- It also prevents the sender from claiming that they never sent the message in the first place (also known as repudiating the message).

Cryptosystems components

Cryptosystems

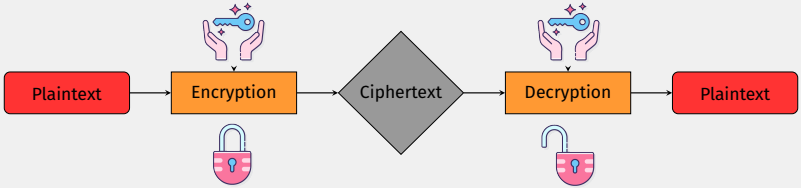
A **cryptosystem** encompasses all of the necessary components for encryption and decryption to take place.

- Software
- Protocols
- Algorithms
- Keys



Modern Cryptographic algorithms

- Modern Cryptographic algorithms are built around complex mathematical functions that combines various operations and methods of how the plaintext is processed with various randomness operations to produce a strong cryptography algorithm.
  - Specific properties make them resistant to reversing or solving by means other than the application of the algorithm and key



While everyone knows how to use a open the lock, without the key to unlock the lock, that knowledge is useless.

Cryptographic Systems

- A cryptographic system is characterised along three independent dimensions:
  - The type of operations used for transforming plaintext to ciphertext, which can perform either **Confusion**, **Diffusion**, or both.
    - Substitution: replacing an item with a different item
    - Transposition: changing the order of items
  - The way in which the plaintext is processed
    - block Cipher
    - Stream Cipher
  - The number of keys used:
    - Symmetric- and Asymmetric
- In addition to these dimensions, there is a unique method to hide information called **Steganography** .

Confusion and Diffusion

- Cryptographic algorithms rely on two basic operations to transform and to obscure plaintext messages: confusion and diffusion.
- Confusion occurs when the relationship between the plaintext and the key is so complex that an attacker cannot easily alter the plaintext and analyse the resulting ciphertext to determine the key.
- Diffusion occurs when a change in the plaintext results in multiple, widespread changes throughout the ciphertext.

Confusion and Diffusion in Practice

- **Substitution ciphers** are a classic example of **confusion**. Each letter in the plaintext is replaced by another symbol, obscuring the relationship between the plaintext and the key.
- **Transposition ciphers** are a classic example of **diffusion**. The letters of the plaintext are rearranged, spreading the influence of each character across the ciphertext.
- Modern cryptographic algorithms combine both substitution and transposition to achieve strong confusion and diffusion.

Confusion and Diffusion in Practice- Online Session

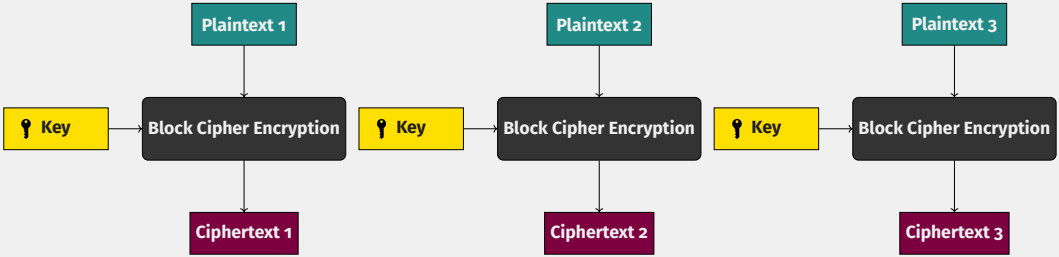
- Practical exercises on classical cipher techniques:
  - Substitution (Caesar cipher)
  - Polyalphabetic substitution
  - Transposition
- These will be covered in an online session:
  - Wednesday 11 February
  - 2:00 pm – 4:00 pm
- The session link will be available on Blackboard

Ciphers

- Cipher systems have long been used by individuals and governments interested in preserving the confidentiality of their communications.
- In addition to how the data is scrambled to ensure secrecy, another dimension of Cryptography focuses on how the data is processed.
- There are two methods in cryptography to process the data in a cryptographic algorithm. The methods are:
  - Block Cipher
  - Stream Cipher
- It's important to remember that these concepts seem somewhat basic, but when used in combination, they can be formidable opponents and cause cryptanalysts many hours of frustration.

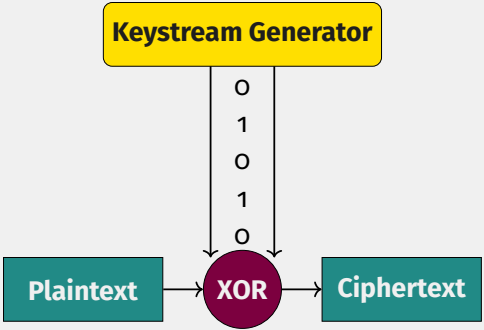
Block Cipher

- Encrypt a block of input to a block of output
- Typically, the two blocks are of the same length
- Most symmetric key systems block size is 64
- In AES block size is 128
- Block cipher has different modes for encrypting plaintext longer than a block
  - Example below is for the Cipher block Chaining (CBC) mode.



Stream Cipher

- A stream cipher does not divide a message into blocks.
- A stream cipher treats the message as a stream of bits and performs mathematical functions on each bit individually.
- Stream ciphers use keystream generators, which produce a stream of bits that is XORed with the plaintext bits to produce ciphertext

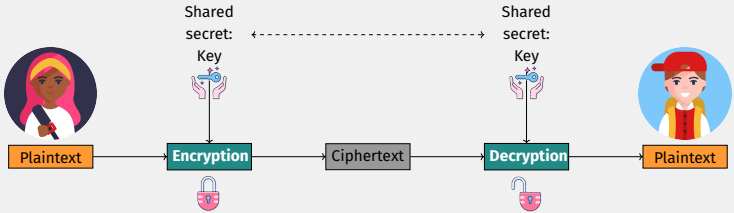


Cryptographic Functions and Ciphers

- Each cipher has specific characteristics that make it desirable or undesirable
- When evaluating a cipher, consider its intended use
  - Are you trying to secure data in transit or data at rest?
  - Different ciphers solve different problems better than others
- After selecting a cipher, you must make additional decisions about key size, operational mode, etc.
- Two types of commonly used encryption algorithms
  - Symmetric algorithms use the same key for encryption and decryption
  - Asymmetric algorithms employ two keys, a public key and a private key, making up what is known as a key pair
- Some type of steganography uses keying to encrypt in addition to hiding data (we will see later)

Symmetric encryptions

- In symmetric cryptography the sender and receiver use two instances of the same key for encryption and decryption
- Symmetric keys are also called shared secret keys.
  - If 100 people were going to communicate, then 4950 pair of keys would be involved.
  - The equation used to calculate the number of symmetric keys needed is $N * (N - 1)/2$ = number of keys
- A simple symmetric encryption example:



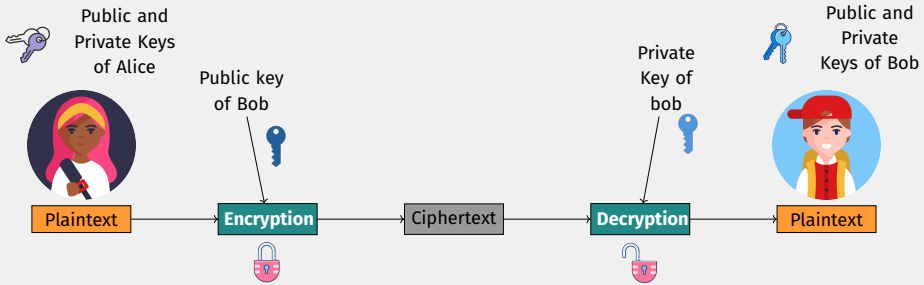
Symmetric encryptions

- Symmetric encryption relies on each user to keep the key a secret and properly protected.
- It is completely dependent on how well users protect the key.
  - Must be a mechanism for key management such as key distribution and key revocation.
- If an intruder were to get this key, they could decrypt any intercepted message encrypted with it.
- Symmetric cryptosystems can provide Confidentiality:
- Symmetric cryptosystems cannot provide authentication or Non-Repudiation.
  - There is no way to prove through cryptography who actually sent a message if two people are using the same key.

Asymmetric encryptions

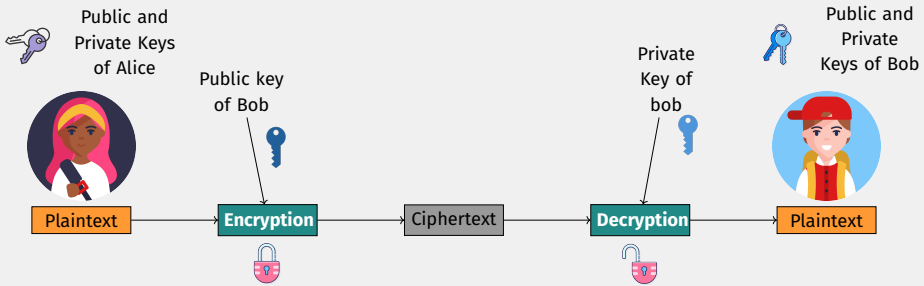
- In Asymmetric encryptions sometimes called public key systems each user has two keys:
  - The public key can be known to everyone, and the private key must be known and used only by the owner.
  - The two different asymmetric keys are mathematically related.
    - In other words, If a message is encrypted by one key, the other key is required in order to decrypt the message.
- Each user using an asymmetric encryption algorithm needs their own pair of keys containing one public key and one private key.
  - If 100 people were going to communicate, then 200 keys or 100 pair of keys would be involved.
- Asymmetric systems also provide for easier and more manageable key distribution than symmetric systems and do not have the scalability issues of symmetric systems.

Asymmetric encryption functions



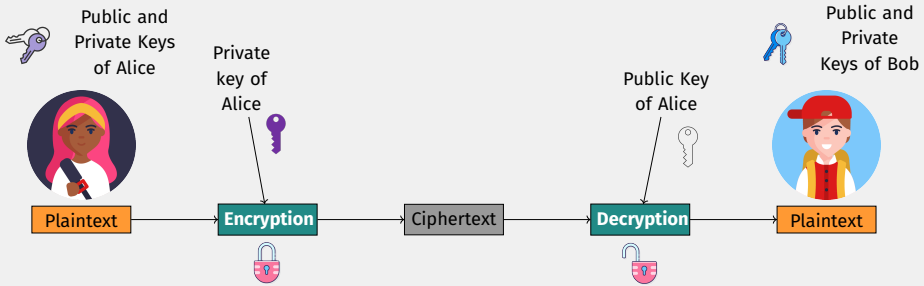
- How **Confidentiality** is provided in asymmetric encryption:
  - If confidentiality is needed Alice would encrypt the file with the receiver's public key.
  - This is called a secure message format because it can only be decrypted by the person who has the corresponding private key.

Asymmetric Encryption Functions



- How **Authentication** is provided in asymmetric encryption:
  - A message can be decrypted with a public key only if the message was encrypted with the corresponding private key.
  - Bob is the only one who is supposed to have his private key. This provides Authentication.

Asymmetric Encryption Functions



- How the **Non-repudiation** function provided in asymmetric encryption:
  - If Alice encrypt the message with her private key.
  - Only her public key can decrypt it.
  - Alice cannot deny sending this message if her public key can decrypt it. This is non repudiation

Symmetric vs. Asymmetric

	Symmetric Encryption	Asymmetric Encryption
Speed	Very fast	Much slower
Strength	Strong with large key sizes	Strong with large key sizes
Scalability	Poor (many keys needed)	Better (fewer keys needed)
Functions	Confidentiality only	Confidentiality, authentication, non-repudiation
Key Distribution	Needs secure key exchange	Uses centralised mechanisms

Steganography

- Steganography hides data inside other media so its existence is concealed (security through obscurity).
- The message is hidden, not encrypted.
- Only the sender and receiver know the message exists.
- Large media files are ideal for hiding data.
- Ordinary images attract less attention than encrypted messages.

Detecting Steganography

- Steganography is often linked to illegal use (e.g. covert communication).
- Detection is difficult because messages attract little attention.
- Hidden data is often encrypted (typically with a symmetric cipher).
- Tools exist to detect steganography (e.g. Stegdetect, StegSpy).

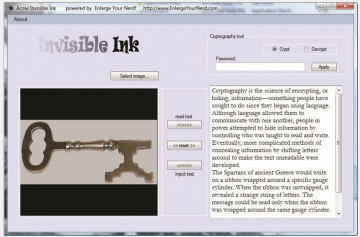


Figure 1: Steganography detection by Invisible Ink tool

Hashing Functions

- A hash (or hashing) function is a special mathematical function that performs a **one-way function**
  - Once the algorithm is processed, there is no feasible way to use the ciphertext to retrieve the plaintext
  - There is no feasible way to generate two different plaintexts that compute to the same hash value
- it is used in cryptography to:
  - Store computer passwords
  - Ensure message integrity
- Hashing produces a unique value
  - Corresponds to the data entered
  - The hash output , called **Message digest** or the **Hash**, is also reproducible by anyone else running the same algorithm against the same data

Hashing Functions

- A hash function H accepts a variable-length block of data x as input and produces a fixed-size hash value:
  - Principal object is data integrity.
- Cryptographic hash function is an algorithm for which it is computationally infeasible to find either:
  a One-way property: A data object that maps to a pre-specified hash result.

    x is the preimage of h for a hash value $h = H(x)$

  b Collision-free property: Two data objects that map to the same hash result

    Collision Occurs if we have $x \neq y$ and $H(x) = H(y)$

Because of these characteristics, hash functions are often used to determine whether or not data has changed.

Hash Function and Integrity

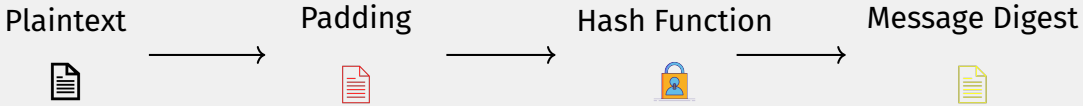
- Hash functions:
  - Help detect forgeries
  - Compute a checksum of a message
  - Combine the checksum with a cryptographic function so that the result is tamper-proof
- A hash is:
  - A checksum designed so that no one can forge a message in a way that will result in the same hash as a legitimate message
  - Usually a fixed size, resulting in a hash value, which is larger than checksum values

Hash Function and Integrity

- The most common hashing functions create the message digest for digitally signed messages
- Hashing is also used to protect user passwords
- Hashing-type functions can also be used with symmetric key cryptography, and the result of the operation is called a message authentication code (MAC)

Padding in Hashing Functions

- Padding is the process of adding extra bits to a message before applying the hash function to ensure the input meets specific requirements for length or format.



Message Authentication Code (MAC)

- Also known as a keyed hash function
- Used by two parties sharing a secret key to authenticate information exchanged between those parties
- MACs are often generated using either HMAC or CMAC

Takes as input a secret key and a data block and produces a hash value (MAC) which is associated with the protected message

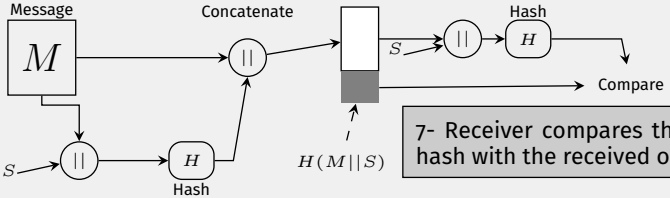
- If the integrity of the message needs to be checked, the MAC function can be applied to the message and the result compared with the associated MAC value
- An attacker who alters the message will be unable to alter the associated MAC value without knowledge of the secret key

Hash-based Message Authentication Code (HMAC)

1- Sender creates the message

4- Concatenate message with hash value

5- Receiver concatenates the message with the secret value



6- Receiver hashes the message–secret pair

7- Receiver compares the computed hash with the received one

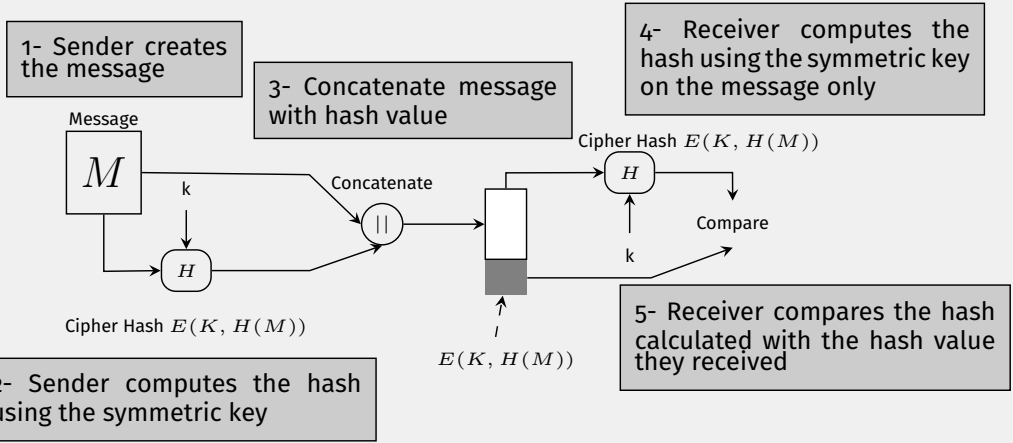
2- Sender concatenates the message with the secret value

3- Sender hashes the message–secret pair

"A" hashes the message with a shared secret and appends the result. The receiver verifies it using the same secret, preventing message tampering.

Ensures authentication, and integrity.

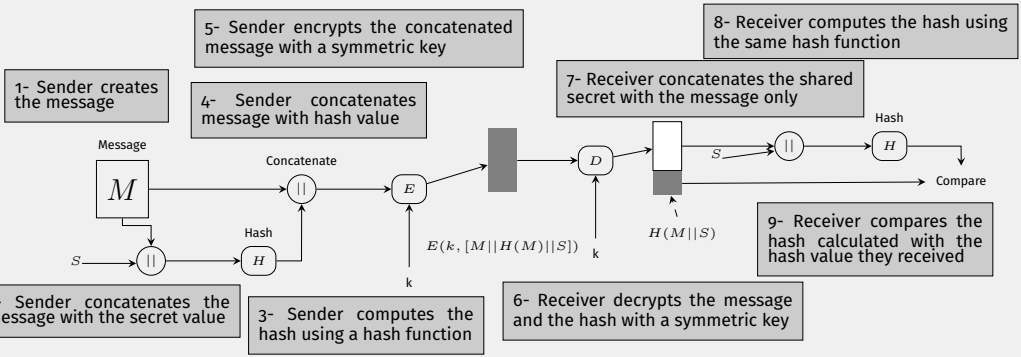
Cipher-based Message Authentication Code (CMAC)



The hash code is computed using symmetric encryption, reducing processing overhead when confidentiality is not required.

Ensures authentication, and integrity.

Authenticated Encryption

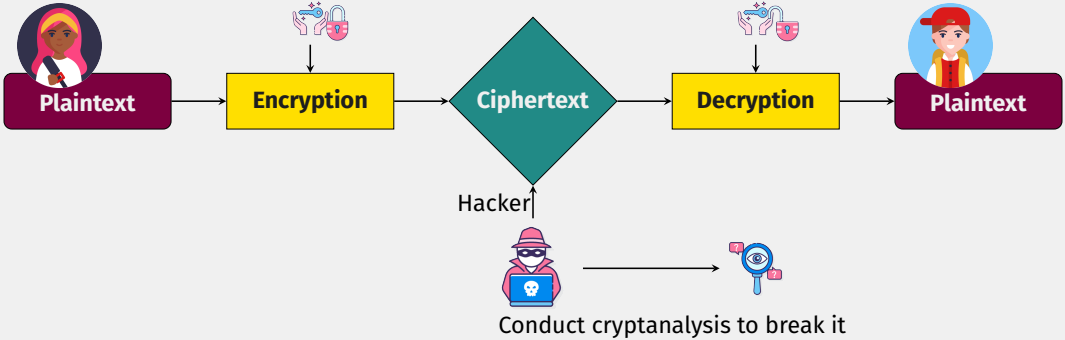


Combines both HMAC and CMAC. Confidentiality can be added by encrypting the message and its hash using a symmetric key and a hash function such as SHA.

Ensures confidentiality, authentication, and integrity.

Cryptanalysis

- Cryptanalysis analyses encrypted data to recover the original message.
- The goal is to break encryption without knowing the secret key.
- It is used by both attackers and security professionals.



Cryptanalysis methods

- **Differential cryptanalysis**: Compares plaintext and ciphertext
- **Collision attack**: Different inputs, same output
- **Brute force**: Tries all possible keys
- **Dictionary attack**: Uses common passwords
- **Frequency analysis**: Studies character patterns
- **Downgrade attack**: Forces weaker security
- **Known plaintext**: Uses known plaintext–ciphertext pairs
- **Quantum computing**: Future threat to cryptography

References

- The lecture notes and contents were compiled from my own notes and from various sources.
- **The lecture notes are very detailed. If you attend the lecture, you should be able to understand the topics.**
  - Chapter 2- Cryptographic Concepts, Hacker Techniques, Tools, and Incident Handling
  - Chapter 8- Encryption, Computer Security Fundamentals
  - Chapter 7- Cryptography, Fundamentals of Information Systems Security
- **Recommended Readings note:** Focus on what was covered in the class.